Chapter 11. Resampling Methods

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1 Training Error and Test Error

Resampling methods repeatedly draw samples from the data at hand to learn about an estimator. The two most common are **cross-validation** and the **bootstrap**; they estimate the prediction error, the standard deviation and the bias of estimates.

Definition 11.1

Let $D_n = \{(x_i, y_i)\}_{i=1}^n$ be the training data, g_{D_n} a prediction rule fitted to it, and L a loss function.

- The **training error** applies the fitted rule back to the training observations, e.g. $\frac{1}{n}\sum (y_i - \hat{y}_i)^2$.
- The **test error** (prediction error) is the expected loss on a new observation (x_0, y_0) independent of D_n:

$$PE = \mathrm{E}[L(y_0, g_{D_n}(x_0))].$$

The training error is easy to compute but usually underestimates the test error, because the rule was tuned to the same data: a curve that zig-zags through every training point has zero training error yet predicts new points $y_i^* = f(x_i) + \varepsilon_i^*$ badly. The smaller the test error, the more suitable the model. Measures such as $MS_E = \frac{SS_E}{n-2}$ describe goodness of fit, not predictive ability.

With an independent test set $\{(x_j, y_j)\}_{j=1}^{n_t}$ we could estimate

$$\widehat{PE} = \frac{1}{n_t}\sum_{j=1}^{n_t} L(y_j, \hat{y}_j), \qquad \hat{y}_j = g_{D_n}(x_j),$$

but a large test set is often unavailable. The methods below hold out part of the training data instead.

2 Cross-Validation

2.1 K-fold cross-validation

Randomly divide the data into K roughly equal parts $C_1, \dots, C_K$ (C_k holds the indices of part k, n_k observations; $n_k = n/K$ if n is a multiple of K). For each k, fit the model to the other $K-1$ parts and predict part k:

$$\mathrm{MSE}_k = \frac{1}{n_k} \sum_{i \in C_k} (y_i - \hat{y}_i)^2, \qquad CV_{(K)} = \sum_{k=1}^{K} \frac{n_k}{n} \mathrm{MSE}_k,$$

where $\hat{y}_i$ is the fit for observation i obtained with part k removed.

2.2 Leave-one-out cross-validation

$K = n$ gives LOOCV. Let $D_{(i)}$ be the training set with (x_i, y_i) omitted and $\hat{y}_{(i)} = g_{D_{(i)}}(x_i)$. Then

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} L(y_i, \hat{y}_{(i)}),$$

which mimics $\widehat{PE}$ by letting each point serve as a test point for a rule that did not see it.

Advantages of LOOCV over a single validation split:

(i) much less bias, since each fit uses $n-1$ observations, almost the whole data set;

(ii) no randomness, since repeated runs give the same result.

Theorem 11.2: LOOCV shortcut for least squares

For least-squares linear regression $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$,

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{y_i - \hat{y}_i}{1 - h_{ii}} \right)^2,$$

where $\hat{y}_i$ is the fitted value from the full fit and h_{ii} is the leverage of observation i. The terms $(y_i - \hat{y}_i)/(1 - h_{ii}) = y_i - \hat{y}_{(i)}$ are the **PRESS residuals**; only one fit is needed.

Derivation: proof

Let $\mathbf{x}_i^\top$ be the ith row of $\mathbf{X}$, $A = \mathbf{X}^\top \mathbf{X}$, and $h_{ii} = \mathbf{x}_i^\top A^{-1} \mathbf{x}_i$. Deleting row i gives $A_{(i)} = A - \mathbf{x}_i \mathbf{x}_i^\top$ and, by the Sherman–Morrison formula,

$$A_{(i)}^{-1} = A^{-1} + \frac{A^{-1} \mathbf{x}_i \mathbf{x}_i^\top A^{-1}}{1 - h_{ii}}.$$

With $\mathbf{X}_{(i)}^\top \mathbf{y}_{(i)} = \mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i$,

$$\hat{y}_{(i)} = \mathbf{x}_i^\top A_{(i)}^{-1} (\mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i) = \left(\mathbf{x}_i^\top A^{-1} + \frac{h_{ii} \mathbf{x}_i^\top A^{-1}}{1 - h_{ii}} \right) (\mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i) = \frac{\mathbf{x}_i^\top A^{-1} (\mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i)}{1 - h_{ii}}$$
$$= \frac{\hat{y}_i - h_{ii} y_i}{1 - h_{ii}}.$$

Hence $y_i - \hat{y}_{(i)} = \dfrac{(1 - h_{ii}) y_i - \hat{y}_i + h_{ii} y_i}{1 - h_{ii}} = \dfrac{y_i - \hat{y}_i}{1 - h_{ii}}$.

Remark 11.3: Choice of K

LOOCV does not shake up the data enough: the n fits are trained on nearly identical sets, so their errors are highly correlated and their average can have high variance. $K = 5$ or 10 is usually a better choice.

2.3 Why cross-validation works: model selection

Example 11.4: Choosing the degree of a polynomial

Consider $y_i = f(x_i) + \varepsilon_i$ fitted by $f_p(x) = \beta_0 + \beta_1 x + \cdots + \beta_p x^p$, and let $\hat{f}_p$ be its LS estimate. The goal is to select p. A natural target is

$$\mathrm{MSE}(p) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}\big(f(x_i) - \hat{f}_p(x_i)\big)^2,$$

but f is unknown. Consider instead the prediction squared error for new responses $y_i^* = f(x_i) + \varepsilon_i^*$:

$$\begin{aligned} PSE(p) &= \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}\big(y_i^* - \hat{f}_p(x_i)\big)^2 = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}\big[(f(x_i) - \hat{f}_p(x_i)) + \varepsilon_i^*\big]^2 \\ &= \mathrm{MSE}(p) + \sigma^2, \end{aligned}$$

because ε_i^* is independent of $\hat{f}_p$ and has mean zero. So minimizing PSE is equivalent to minimizing MSE. Now let

$$CV(p) = \frac{1}{n}\sum_{i=1}^{n} \big(y_i - \hat{f}_p^{-i}(x_i)\big)^2,$$

where $\hat{f}_p^{-i}$ is fitted without the ith point. Since y_i is independent of $\hat{f}_p^{-i}$,

$$\mathrm{E}\big(y_i - \hat{f}_p^{-i}(x_i)\big)^2 = \sigma^2 + \mathrm{E}\big(f(x_i) - \hat{f}_p^{-i}(x_i)\big)^2,$$

and if $\hat{f}_p^{-i} \approx \hat{f}_p$ (leaving out one point changes little), $\mathrm{E}[CV(p)] \approx \sigma^2 + \mathrm{MSE}(p) = PSE(p)$. Thus $\hat{p} = \arg\min_p CV(p)$ is a sensible rule. The training error $\frac{1}{n}\sum(y_i - \hat{f}_p(x_i))^2$ does not have this property: it keeps decreasing as p grows.

3 The Bootstrap

The bootstrap quantifies the uncertainty of an estimator or a learning method, e.g. the standard error of a coefficient or a confidence interval for it.

3.1 A simple example

We invest a fraction α of our money in an asset with return X and $1-\alpha$ in one with return Y, and choose α to minimize the risk $\mathrm{Var}(\alpha X + (1-\alpha)Y)$.

Theorem 11.5

The risk is minimized by

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$

Proof. $g(\alpha) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}$, and

$$g'(\alpha) = 2\alpha\sigma_X^2 - 2(1-\alpha)\sigma_Y^2 + 2(1-2\alpha)\sigma_{XY} = 0 \iff \alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) = \sigma_Y^2 - \sigma_{XY}.$$

Since $g'' = 2\mathrm{Var}(X - Y) \geq 0$, this is a minimum. ■

The variances are unknown, so we plug in sample estimates to get $\hat{\alpha}$. To learn the standard deviation of $\hat{\alpha}$ in a simulation, we could draw 1,000 independent data sets of 100 pairs from the population, compute $\hat{\alpha}_1, \ldots, \hat{\alpha}_{1000}$, and report

$$\bar{\alpha} = \frac{1}{1000}\sum_{r=1}^{1000} \hat{\alpha}_r, \qquad \sqrt{\frac{1}{1000-1}\sum_{r=1}^{1000}(\hat{\alpha}_r - \bar{\alpha})^2}.$$

3.2 Back to the real world

For real data we cannot draw new samples from the population. The bootstrap instead draws data sets *from the original data set with replacement*, using the computer to mimic the process of generating new samples.

Definition 11.6: Bootstrap standard error

1. From the original data Z of size n, draw a bootstrap data set Z^{*1} of size n by sampling observations with replacement (some observations appear more than once, some not at all).
2. Compute the estimate $\hat{\alpha}^{*1}$ from Z^{*1}.
3. Repeat B times (say $B = 100$ or 1000) to get $\hat{\alpha}^{*1}, \ldots, \hat{\alpha}^{*B}$.
4. Estimate the standard error by

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \bar{\alpha}^*\right)^2}, \qquad \bar{\alpha}^* = \frac{1}{B}\sum_{r'=1}^{B}\hat{\alpha}^{*r'}.$$

Example 11.7: Three observations

With $Z = \{(4.3, 2.4), (2.1, 1.1), (5.3, 2.8)\}$ (observations 1, 2, 3), possible bootstrap samples are $Z^{*1} = \{3, 1, 3\}$, $Z^{*2} = \{2, 3, 1\}$, $\ldots$, $Z^{*B} = \{2, 2, 1\}$ (by observation number), each yielding its own $\hat{\alpha}^{*r}$.

In R, `sample(x, replace=T)` draws a bootstrap sample of the indices.